

Phase 3: Orthogonal Refraction and Multi-Dimensional Reflection

Analyze your cognitive progress throughout this activity by drafting a structured meta-cognitive reflection that directly addresses the following prompts:

1. **The Conceptual Evolution of Multiplication:** Contrast your past understanding of multiplication in scalar mathematics with the geometric reality of the 3D dot product. How does the dot product serve as an indicator of directional alignment? Explain how the transition from the algebraic component formula to the geometric cosine formula alters your spatial understanding of vector interactions.
2. **Visualizing Projections and Decompositions:** Detail the specific visual or structural hurdles you encountered when projecting one vector onto another within a three-dimensional coordinate grid. How did you maintain clarity between scalar components and vector projections? What geometric strategies (such as tracking 2D planar cross-sections) did you employ to ensure your algebraic calculations matched your physical intuition of a "shadow"?

Read the rubric before beginning to identify the Gold Standard!!!

1. Before learning about the dot product, I thought vector multiplication was simply a scalar operation. Multiplying two numbers usually simply yields another number. I tend to focus more on the calculation process itself, without considering geometric information such as direction. After learning about the dot product of the 3D vectors, I realize that dot product actually reflects both the magnitudes of the two vectors and the directional relationship between them.

I first understand dot product by $u \cdot v = u_1v_1 + u_2v_2 + u_3v_3$. It just calculates the final scalar, but does not tell me what the scalar really stands for. Using the equation $u \cdot v = ||u|| ||v|| \cos(\theta)$ I realize that the dot product can actually be used to measure the degree to which two vectors are aligned in direction. This is because $\cos(\theta)$ varies with the angle between them. When two vectors point in the same direction, $\theta = 0$. At this point, $\cos(\theta) = 1$, and the dot product is a large positive value. When two vectors are perpendicular to each other, $\theta = 90$, $\cos(\theta) = 0$, and therefore the dot product is zero. When the angle between the two vectors is greater than 90, the cosine is negative, and the dot product also becomes negative. The first formula tells me how to calculate it. The cosine formula tells me the meaning of the result. The dot product actually describes the spatial relationship between the two vectors themselves, not just the numerical values of their coordinates.

Plus, work can be expressed as $W = F \cdot D$. This shows that it is not the force as a whole that actually does work, but that component of the force which is aligned with the direction of displacement. It makes me understand that the core of the dot product is not ordinary multiplication, but a measure of the extent to which one vector's components align with the direction of another vector.

2. The main difficulty I encountered when studying vector projections in 3D space was not the formulae themselves, but how to link the algebraic calculations to the geometric figures in space. In 2D, projections are easy to visualise because I can directly see the angle and perpendicular lines. In 3D, 2 vectors do not necessarily lie along any of the x, y, or z axes. I found it difficult to determine from the diagram exactly where the projection should be located and how to draw it. I solve the visualisation problem by stopping trying to visualise the entire 3D coordinate system at once, but on the 2D plane defined by the vectors a and b . Within this plane, the projection can become a standard trigonometric problem. I can understand $|b| \cos(\theta)$ as the adjacent side in a right triangle along the direction of a , like the shadow of b on the side a .

At first, I thought that a scalar component and a vector projection were almost the same thing because they both describe a part of a vector in the direction of another vector. Later, I realize that a scalar component is directionless, and a vector projection is directional. The scalar component is $|b| \cos(\theta)$, which represents the length of b in the direction of a . The vector projection is $(a \cdot b) / (|a|^2) * a$. It has not only magnitude but also direction (parallel to a). I interpret the scalar component as the length of the shadow, and the vector projection as both the length and direction of the shadow. This helps me to distinguish between these two concepts.

I have also learnt to check my algebraic calculations using geometric relationships. If $\text{project}_a(b)$ is calculated correctly, then it must be parallel to a .